How the dry years 2003 and 2018 affected Swiss rivers and run-of-river hydropower

Anas Meer Ahmed | BSc Energy and Environmental Systems Engineering, HSLU | November 2026

Summary. I looked at how the dry years 2003 and 2018 affected Swiss rivers and run-of-river hydropower. Small lowland rivers dropped to 10–30 % of their normal flow, while a glacier-fed river stayed close to normal. National run-of-river production dropped at the same time as the rivers, but only to about 75 % of normal.

1. Question

Switzerland gets a large share of its electricity from hydropower, and 2003 and 2018 were both very dry years. I wanted to find out how much river flow dropped in those years, and whether run-of-river power production dropped in the same way.

2. Data

- River discharge: the CAMELS-CH dataset (Höge et al., 2023), with daily discharge for 331 catchments from 1981 to 2020. 140 of these are Swiss river gauges with no missing days. I left out lake gauges and stations with gaps.
- Electricity: the monthly Swiss electricity balance from the Swiss Federal Office of Energy (SFOE). It reports run-of-river production separately from storage hydropower and starts in 2000.

I picked six catchments with no hydropower plants or reservoirs listed in the dataset, covering the main types of Swiss rivers:

River (gauge)	Type	Area (km²)	Mean elevation (m)
Weisse Lütschine (Zweilütschinen)	Glacier-fed, Alpine	165	2,148
Plessur (Chur)	Snowmelt, Alpine	264	1,864
Emme (Emmenmatt)	Pre-Alpine	443	1,071
Lorze (Frauenthal)	Lowland, Central Switzerland	262	681
Töss (Neftenbach)	Lowland, rain-fed	343	658
Broye (Payerne)	Lowland, rain-fed	416	723

3. Method

I averaged the daily discharge to monthly values. For each station and month I calculated a normal value: the 1981–2020 average for that month, without 2003 and 2018, so the dry years would not pull the normal down. I then expressed each month of 2003 and 2018 as a percentage of normal. I did the same for run-of-river production, with 2000–2020 as the baseline because the electricity data starts in 2000. All of the analysis was done in R (tidyverse, ggplot2).

4. Results

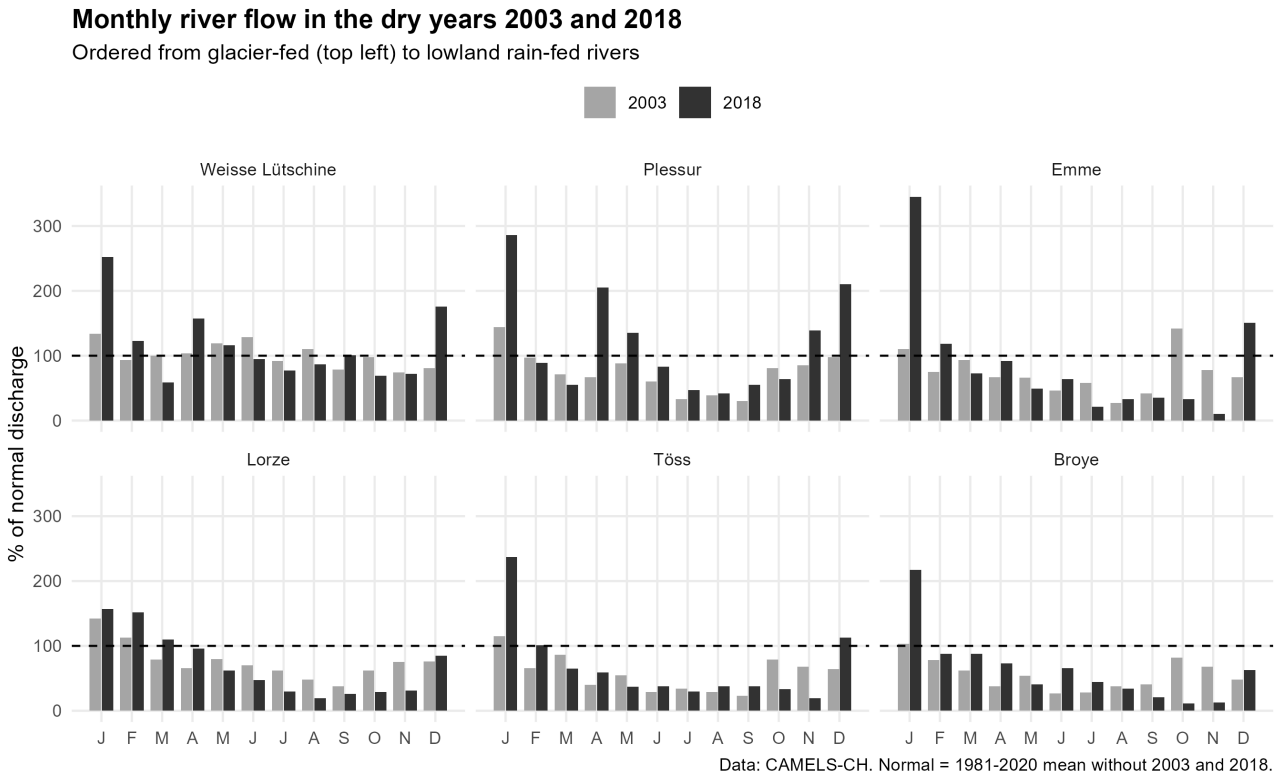


Figure 1. Monthly discharge in 2003 and 2018 as a percentage of normal, for six catchments.

2003 and 2018 were dry in different ways. In 2003 the lowland rivers dropped to 23–28 % of normal between June and September and had mostly recovered by October. 2018 started with a very wet January (157–345 % of normal), but the dry period lasted longer and was worst in autumn. The Broye was at 11 % of normal in October.

The lowland rivers were hit hardest. In 2018 the four lowland and pre-Alpine rivers averaged 18–31 % of normal from July to November.

The glacier-fed Weisse Lütschine hardly dropped at all. It was above normal in summer 2003 (110–129 %) and close to normal in 2018 (77–101 % from July to September). The heat made the glacier melt more, which made up for the missing rain.

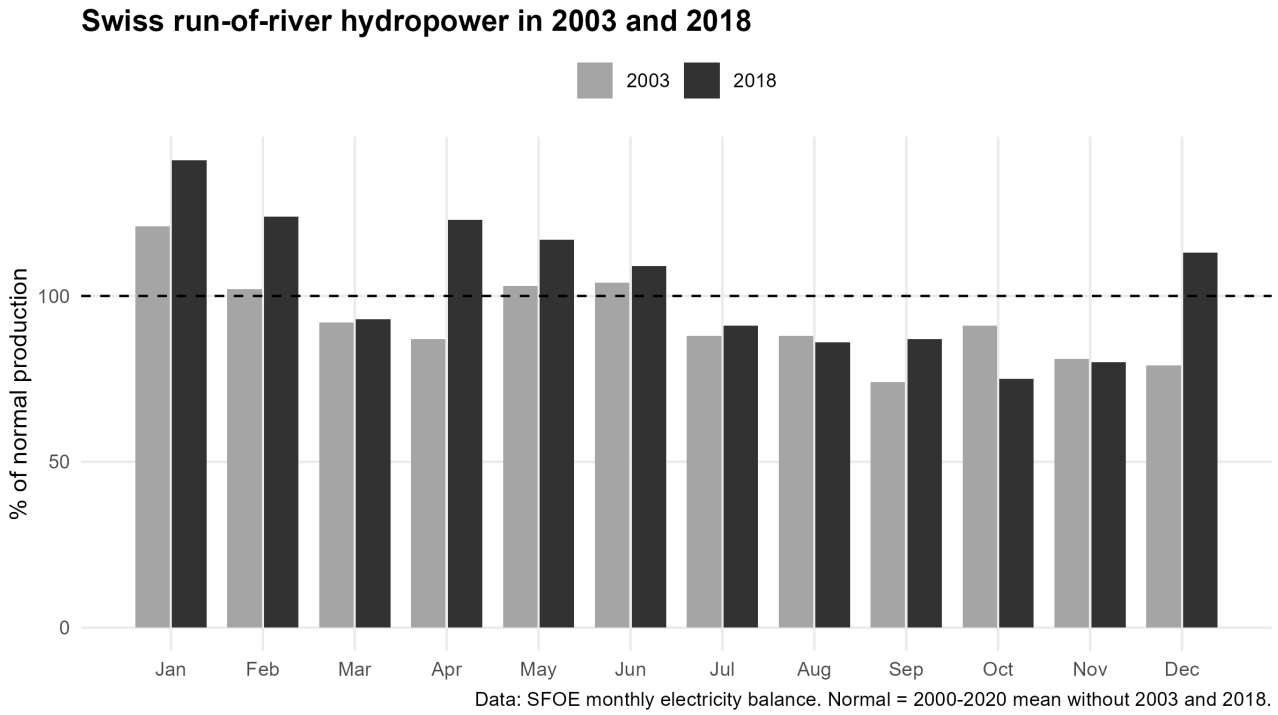


Figure 2. Swiss run-of-river production in 2003 and 2018 as a percentage of normal.

Run-of-river production dropped at the same time as the rivers. The lowest month was September in 2003 (74 % of normal) and October in 2018 (75 %). The drop was much smaller than in the lowland rivers, though.

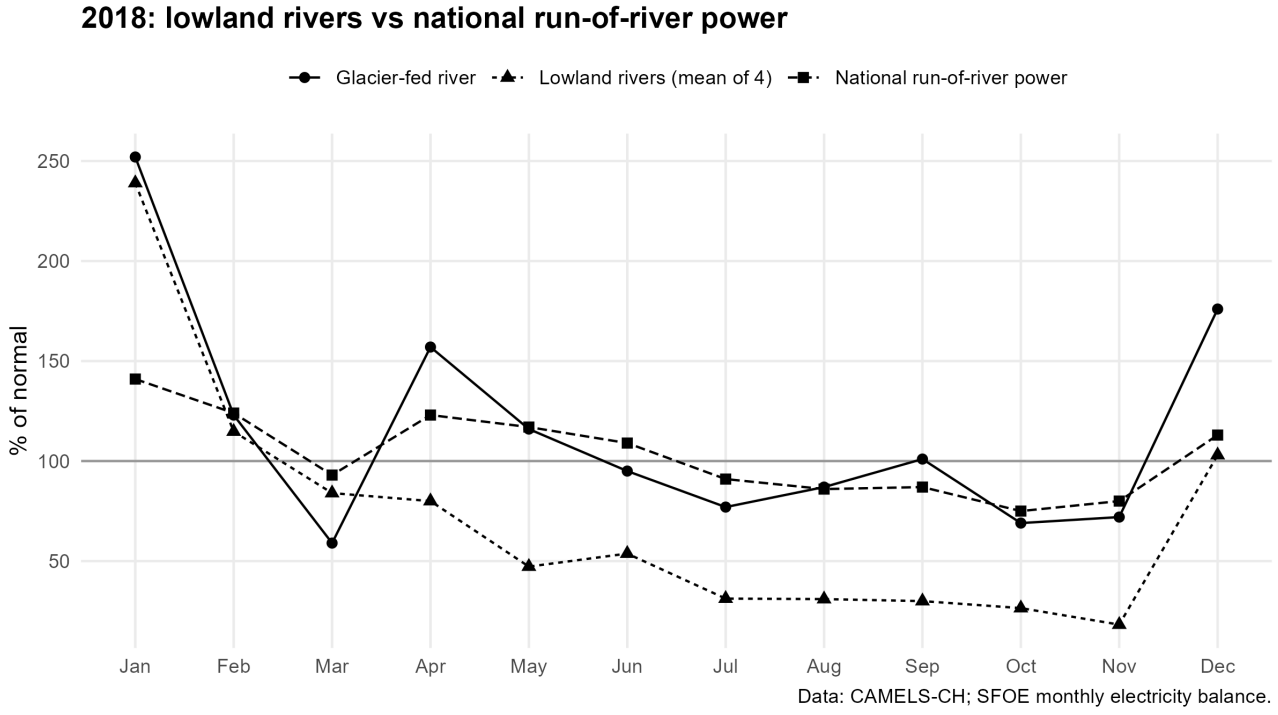


Figure 3. 2018: mean of four lowland rivers, the glacier-fed river and national run-of-river production, as a percentage of normal.

In 2018, national production followed the glacier-fed river much more closely than the lowland rivers. I think this is because most run-of-river power comes from plants on big rivers like the Rhine, Aare and Rhône. They get their water from Alpine snow, glaciers and large lakes, so their flow drops less in a dry year. I have not tested this here (see limitations).

5. Limitations

- Six catchments are not enough to represent all of Switzerland. Also, having no hydropower plants listed in the dataset does not mean the flow is completely natural.
- More run-of-river plants were built between 2000 and 2020. Using one fixed baseline makes 2018 look slightly better than it really was.
- I did not test why production dropped less than the rivers. For that I would need discharge data for the big rivers where the main plants are.
- Monthly averages hide short extreme periods. I also left out storage hydropower, because its output depends more on electricity prices than on how much water is flowing in.

6. Conclusion

In dry years, small lowland rivers can lose 70–90 % of their normal flow, while glacier-fed rivers are protected by melt. National run-of-river production dropped by about a quarter at most. Glaciers are shrinking, so this protection will get smaller over time. A next step would be to look at how the big rivers respond in dry years.

References

Höge, M., Kauzlaric, M., Siber, R., et al. (2023). CAMELS-CH: hydro-meteorological time series and landscape attributes for 331 catchments in hydrologic Switzerland. *Earth System Science Data*, 15, 5755–5784. https://doi.org/10.5194/essd-15-5755-2023

Swiss Federal Office of Energy (SFOE). Swiss electricity balance, monthly values (ogd35). opendata.swiss.